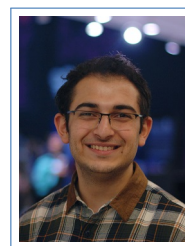


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Education

- 2020–2025 **University of British Columbia**, Vancouver, Canada
Ph.D. in Computer Science, Advisors: Anne Condon and Khanh Dao Duc
Dissertation Title: Optimal Transport Algorithms and Metrics for 3D Point-Cloud Alignment with Applications to Cryogenic Electron Microscopy
- 2015–2020 **Sharif University of Technology**, Tehran, Iran
B.Sc. in Computer Engineering, minor in Mathematics GPA:18.16/20

Publications

Submitted/Preprints

- 2025 **The Joint Gromov-Wasserstein Objective for Multiple Object Matching**, *A Tajmir Riahi, K Dao Duc*, available on arXiv preprint arXiv:2511.16868.

Published in peer-reviewed journals

- 2025 **Alignment of Partially Overlapping Cryo-EM Maps Using Unbalanced Gromov-Wasserstein Divergence**, *A Tajmir Riahi, C Zhang, A Condon, J Chen, K Dao Duc*, PRX Life, 3(2), 023003.
- 2023 **AlignOT: An Optimal Transport Based Algorithm for Fast 3D Alignment With Applications to Cryogenic Electron Microscopy Density Maps**, *A Tajmir Riahi, G Woollard, F Poitevin, A Condon, K Dao Duc*, IEEE/ACM Transactions on Computational Biology and Bioinformatics, 20(6), 3842-3850.

Presented in peer-reviewed workshops

- 2023 **EMPOT: Partial Alignment of Density Maps and Rigid Body Fitting using Unbalanced Gromov-Wasserstein Divergence**, *A Tajmir Riahi, C Zhang, J Chen, A Condon, K Dao Duc*, Machine Learning in Structural Biology Workshop at the 37th Conference on Neural Information Processing Systems
- 2022 **3D Alignment of Cryogenic Electron Microscopy Density Maps by Minimizing Their Wasserstein distance**, *A Tajmir Riahi, G Woollard, F Poitevin, A Condon, K Dao Duc*, Machine Learning in Structural Biology Workshop at the 36th Conference on Neural Information Processing Systems

Presentations

- Jun 2024 **Summer School on Optimal Transport, Stochastic Analysis and Applications to Machine Learning**, *Daejeon, South Korea*
- Mar 2024 **UBC Math Bio Seminar**, *Vancouver, Canada*
- Sep 2023 **Workshop on Mathematical Methods for Exploring and Analyzing Morphological Shapes across Biological Scales**, *Banff, Canada*
- May 2022 **PIMS workshop on Mathematical and Computational Challenges in Cryogenic Electron Microscopy (Cryo-EM)**, *Vancouver, Canada*

Softwares

OTAlignment ChimeraX Bundle, *Implemented all methods used in my PhD, including AlignOT, EMPOT, JGW for alignment, and CVT, TRN for point sampling as a ChimeraX bundle.*

Teaching Exprience

Teaching Assistant

- Spring 2025 **Intermediate Algorithm Design and Analysis**, *CPSC 320 UBC*, (Susanne Bradley, Alan Hu)
- Fall 2024 **Intermediate Algorithm Design and Analysis**, *CPSC 320 UBC*, (Susanne Bradley, Anne Condon)
- Spring 2024 **Intermediate Algorithm Design and Analysis**, *CPSC 320 UBC*, (Susanne Bradley, Bruce Shepherd)
- Spring 2023 **Intermediate Algorithm Design and Analysis**, *CPSC 320 UBC*, (Alan Hu)
- Fall 2022 **Intermediate Algorithm Design and Analysis**, *CPSC 320 UBC*, (Patrice Belleville)
- Spring 2022 **Intermediate Algorithm Design and Analysis**, *CPSC 320 UBC*, (Patrice Belleville)
- Fall 2019 **Design Algorithms**, (Hamid Zarrabi-zadeh)
- Fall 2019 **Data Structures and Algorithms**, (Masoud Seddighin)
- Spring 2019 **Discrete Structures**, *Sharif UT*, (Hamid Zarrabi-zadeh)
- Spring 2019 **Algorithmic Game Theory**, *Sharif UT*, (Masoud Seddighin)
- Fall 2018 **Design Algorithms**, *Sharif UT*, (Hamid Zarrabi-zadeh)
- Spring 2018 **Data Structures and Algorithms**, *Sharif UT*, (Ali Sharifi-Zarchi)
- Spring 2018 **Artificial Intelligence**, *Sharif UT*, (Ali Feyzbakhsh)
- Spring 2018 **Theory of Languages and Automata**, *Sharif UT*, (Ali Movaghar)
- Fall 2017 **Probability and Statistics**, *Sharif UT*, (Abolfazl Motahari)
- Fall 2016 **Fundamentals of Programming**, *Sharif UT*, (Omid Gheibi)

Volunteer Exprience

Technical team member, Sharif AI Challenge 2018

A technical team member in Sharif AI Challenge held by the Students Scientific Chapter of Computer Engineering Department in Sharif University. Website of the event: aichallenge.sharif.edu

Technical team member, Webelopers 2018

A technical team member in the two day event held by the Students Scientific Chapter of Computer Engineering Department in Sharif University.

Scientific-head and co-organizer, Karsoogh 2017, 2018 and 2019

Scientific-head and teacher in a national annual workshop on mathematics for junior high school students, which consists of two entrance exams and a three 24hours day event in August.

Honors and Awards

- 2019 *Ranked 7th in the Nationwide University M. Sc. Entrance Exam for Artificial Intelligence*
- 2019 *Ranked 5th in the National Olympiad of Computer Engineering*
- 2015-2019 *Iran's National Elites Foundation Fellowship*
- 2014 *Gold Medal in National Mathematical Olympiad*

Possible References

Dr. Anne Condon, Professor, Computer Science, UBC

Dr. Khanh Dao Duc, Assistant Professor, Mathematics, UBC

Dr. Young-Heon Kim, Professor, Mathematics, UBC

Dr. Bruce Shepherd, Professor, Computer Science, UBC

Dr. Manu Madhav, Assistant Professor, Biomedical Engineering, UBC